

PlantDeepSUMO

A Deep Learning-Based Webserver for Predicting Lysine SUMOylation Sites in Plants

Quick Start Guide

Webserver: <https://jasonxu.shinyapps.io/PlantDeepSUMO/>

Source Code: <https://github.com/Jasonxu0109/PlantDeepSUMO>

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Overview

PlantDeepSUMO is a CNN-based deep learning tool that predicts lysine SUMOylation sites in plant proteins. It achieves an AUC-ROC of 0.93 and MCC of 0.75 on independent validation data, significantly outperforming existing tools. The webserver supports:

- Single & batch SUMOylation site prediction from protein sequences
- Interactive multiple sequence alignment viewer (SUMOymosa) for predicted sites

Quick Start: SUMOylation Site Prediction

Step 1: Open the Webserver

PlantDeepSUMO

PlantDeepSUMO SUMOymosa

This tool scans for SUMOylation sites from your input sequences.

Paste input FASTA sequence(s) (e.g., 10 sequences) in the area below:

```
>AL6
MEGITHPIPTVEEVFSDFGRRAGLIKALTNDMVKFYQTCDEKENLCLYGLPNETWEV
NLPVEEVPPPELPEPALGINFARDGMQEKDWWVSLVAVHSDSWLLSVAFYFGARFGFGKNER
KRLFQMINELPTIFEVVSGNAKQSKDLSVNNNNNSKSPSGVKSRQSESLSKVAKMSSPPP
KEEEEEDESEDESEDDEQGAVCAGCGDNYGTDEFWICCDACEKWFGKCVKITPAKAEH
IKHYKCPTCSNKRARP
```

Or, upload file (<1 Mb)

Browse Format: FASTA

Submit Download Results (CSV)

Navigate to <https://jasonxu.shinyapps.io/PlantDeepSUMO/> in your web browser. The prediction interface will load automatically.

Step 2: Input Your Protein Sequence

PlantDeepSUMO
PlantDeepSUMO
SUMOymsa

This tool scans for SUMOylation sites from your input sequences.

Paste input FASTA sequence(s) (e.g., 10 sequences) in the area below:

>AL6
MEGITHPIPTVEEVFSDFRGRRAGLIKALTNDMVKFYQTCDPEKENLCLYGLPNETWEV
NLPVEEVPELPEPALGINFARDGMQEKDWVSLVAVHSDSWLLSVAFYFGARFGFGKNER
KRLFQMINEPLTIFEVVSGNAKQSKDLSVNNNNNSKSPSGVKSRSQSELSKVAKMSSPPP
KEEEEEDESEDESEDDEQAVCGACGDNYGTDEFWICCDACEKWFHGKCVKITPAKAEH
IKHYKCPTCSNKRARP

paste protein sequence

Or, upload file (<1 Mb)

Browse

Format: FASTA

upload a protein sequence file

Submit

Download Results (CSV)

You can provide protein sequences in FASTA format using either method:

- **Paste directly:** Type or paste your FASTA-formatted sequence(s) into the input text box.
- **Upload a file:** Click the "Upload" button to select a FASTA file from your computer.

Example FASTA format:

```
>AT1G01010 | Arabidopsis thaliana protein
MDPNVVVDFSTANDKSVERRNVNKLTAYSKGKI...
```

Tip: For batch prediction, include multiple FASTA entries in one submission. The server handles files up to ~1 MB (~2,000 sequences).

Step 3: Submit and Wait for Results

PlantDeepSUMO
PlantDeepSUMO
SUMOymsa

This tool scans for SUMOylation sites from your input sequences.

Paste input FASTA sequence(s) (e.g., 10 sequences) in the area below:

>AL6
MEGITHPIPTVEEVFSDFRGRRAGLIKALTNDMVKFYQTCDPEKENLCLYGLPNETWEV
NLPVEEVPELPEPALGINFARDGMQEKDWVSLVAVHSDSWLLSVAFYFGARFGFGKNER
KRLFQMINEPLTIFEVVSGNAKQSKDLSVNNNNNSKSPSGVKSRSQSELSKVAKMSSPPP
KEEEEEDESEDESEDDEQAVCGACGDNYGTDEFWICCDACEKWFHGKCVKITPAKAEH
IKHYKCPTCSNKRARP

Or, upload file (<1 Mb)

Browse

Format: FASTA

Submit

Download Results (CSV)

Click the "Submit" button. Predictions for typical sequences are generated within seconds. For large batch submissions, please allow additional processing time.

Step 4: Review Prediction Results

Submit
Download Results (CSV)

Prediction of SUMOylation sites below:

Show 10 entries

Search:

	Name	Gene	Position	Probability	Confidence	Flanking_sequence
1	AL6:28	AL6	K28	0.921	very high	XXXMEGITHPIPTVEEVFSDFRGRAGLIALTNDMVKFYQTCDPEKENLCLYGLPNETW
2	AL6:36	AL6	K36	0.9731	very high	HPIPTVEEVFSDFRGRAGLIKALTNDMVFYQTCDPEKENLCLYGLPNETWEVNLPEE
3	AL6:45	AL6	K45	0.7808	median	VFSDFRGRAGLIKALTNDMVKFYQTCDEENLCLYGLPNETWEVNLPEEVPPELPEPA
4	AL6:88	AL6	K88	0.9247	very high	WEVNLPEEVPPELPEPALGINFARDGMQEDWVSLVAVHSDSWLLSVAFYFGARFGGKN
5	AL6:117	AL6	K117	0.9759	very high	EKDWVSLVAVHSDSWLLSVAFYFGARFGGNERKRLFQMINELPTIFEVVSNGAKQSKDL
6	AL6:121	AL6	K121	0.9729	very high	VSLVAVHSDSWLLSVAFYFGARFGGKNERRLFQMINELPTIFEVVSNGAKQSKDLSVNN
7	AL6:142	AL6	K142	0.8206	high	RFGFGKNERKRLFQMINELPTIFEVVSNGAKQSKDLSVNNNNNSKSPGVKSRQSESLSKV
8	AL6:145	AL6	K145	0.9722	very high	FGKNERKRLFQMINELPTIFEVVSNGAKQSKDLSVNNNNNSKSPGVKSRQSESLSKVAKM
9	AL6:155	AL6	K155	0.9276	very high	QMINELPTIFEVVSNGAKQSKDLSVNNNNNSKSPGVKSRQSESLSKVAKMSSPPPEEEEE
10	AL6:157	AL6	K157	0.8805	high	INELPTIFEVVSNGAKQSKDLSVNNNNNSKSPGVKSRQSESLSKVAKMSSPPPEEEEE

Showing 1 to 10 of 21 entries

Previous
1
2
3
Next

The results table displays:

- Gene/Protein identifier
- Lysine residue position of the predicted SUMOylation site
- Flanking sequence around the predicted site
- Prediction probability score (0–1)

Click the "Download" button above the table to export results as a CSV file for downstream analysis.

Interpreting Confidence Levels

Prediction probability scores are categorized into four confidence tiers:

Confidence Level	Probability Range	Interpretation
Very High	$0.90 \leq P < 1.00$	Strongly recommended for experimental validation
High	$0.80 \leq P < 0.90$	Reliable predictions; good candidates for validation
Medium	$0.65 \leq P < 0.80$	Moderate confidence; consider flanking context
Low	$0.50 \leq P < 0.65$	Lower confidence; use with caution

SUMOymsa: Sequence Alignment Viewer

The integrated SUMOymsa viewer allows you to visualize sequence conservation around any predicted SUMOylation site. After prediction, click on a specific site in the results table to open the alignment viewer and inspect flanking residue conservation across species.

This tool implements functions to visualize multiple sequence alignments.

Method:

Muscle ▾

Start:

60

End:

80

Upload a file

Upload...

No file selected

Q5Z9E5	N	L	P	K	H	C	D	N	N	E	V	L	K	A	L	C	N	E	A	G	W
Q0EUF1	T	L	P	K	H	C	D	N	N	E	V	L	K	A	L	C	N	E	A	G	W
Q49404	K	L	P	K	H	C	D	N	N	E	V	L	K	A	L	C	N	E	A	G	W
Q9ZV88	E	L	P	K	H	C	D	N	N	E	V	L	K	A	L	C	N	E	A	G	W
GhBE51	N	L	P	K	H	Y	N	N	N	E	V	L	K	A	L	C	A	K	A	G	W
Q8S307	N	L	P	K	H	C	D	N	N	E	V	L	K	A	L	C	V	E	A	G	W
Q9LN63	N	L	P	K	H	C	D	N	N	E	V	L	K	A	L	C	S	E	A	G	W
B8B755	N	L	P	K	H	C	D	N	N	E	V	L	K	A	L	C	R	E	A	G	W
Q7X196	N	L	P	K	H	C	D	N	N	E	V	L	K	A	L	C	R	E	A	G	W
Q94A43	K	L	P	K	H	C	D	N	N	E	V	L	K	A	L	C	L	E	A	G	W
Q9S7F3	K	L	P	K	H	C	D	N	N	E	V	L	K	A	L	C	L	E	A	G	W

60 65 70 75 80

Running Locally (Large Datasets)

For whole-proteome analyses or large-scale batch predictions beyond the web interface capacity, use the local execution scripts:

1. Download the Jupyter notebook from GitHub: [example.ipynb](#)
2. Install the required Python dependencies (TensorFlow/Keras, NumPy, etc.).
3. Run predictions offline using the pre-trained CNN model.

Citation

If you use PlantDeepSUMO in your research, please cite:

Xu, C. PlantDeepSUMO, a deep learning-based webserver to predict lysine SUMOylation sites in plants. (2026).